Nirma University,

Ahmedabad

A project report on

"Analysis of Crime in India (2001 - 2012)"

Submitted to

"Dr. Saurin Parikh"

Sub : Data Analysis and visualization

By

22BCE160-Kshitij Trivedi

22BCE294-Himanshi Rakholiya

22BCE311-Sana Narmawala

Introduction :

Crime against women has been a significant issue worldwide, and India, like many other countries, faces a disproportionate number of crimes targeting women. These crimes range from sexual violence to social abuses, all of which have a long-lasting impact on the physical and mental well-being of women. Among the various types of crimes, rape, kidnapping and abduction, dowry deaths, assault on women, and cruelty by husbands are some of the most severe and socially damaging offenses. Despite various legal measures and government initiatives aimed at curbing such crimes, the rates of offenses against women continue to remain a cause for concern.

This study seeks to explore the relationship between crime rates against women and their occurrence across different states and union territories (UTs) of India. By examining the different types of crimes, this analysis aims to shed light on the regional variations in crime against women and the potential factors that contribute to these differences.

Data Description:

The data provided in this study contains information about various crimes against women, categorized by state/UT and year. The types of crimes included in the dataset are:

1. **Rape**: Incidents of sexual assault involving force or coercion.
2. **Kidnapping and Abduction**: Includes cases of forcibly taking or holding women against their will.
3. **Dowry Deaths**: Deaths resulting from dowry-related violence or harassment.

4. **Assault on Women with Intent to Outrage Her Modesty**: Physical or verbal attacks aimed at violating the dignity of women.
5. **Insult to Modesty of Women**: Acts that humiliate or degrade women in public or private settings.
6. **Cruelty by Husband or His Relatives**: Emotional, physical, or mental abuse committed by the husband or in-laws.
7. **Importation of Girls**: The illegal trade or transportation of women, often for exploitative purposes.
8. **Total number of persons under trial during the year**: The total count of individuals undergoing judicial proceedings within a specific year.
9. **Persons convicted**: The number of individuals found guilty of charges and sentenced during the year.
10. **Persons acquitted**: The number of individuals found not guilty and released from charges during the year.

The dataset is organized by the year of occurrence and the specific state/UT, enabling the analysis of trends and the geographical distribution of these crimes.

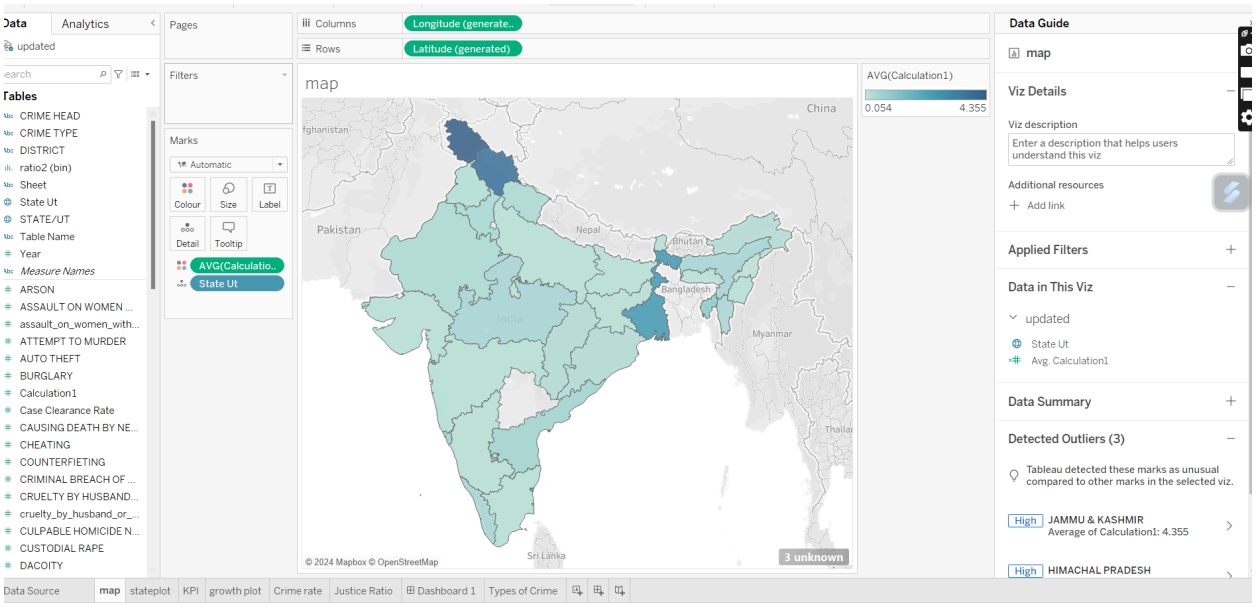
Objective:

The primary objectives of this analysis are:

1. **To analyze the trends in crimes against women**: Identifying how crime rates fluctuate over the years and the specific types of crimes that are more prevalent in certain regions.
2. **To assess regional variations**: Understanding how different states and UTs in India exhibit varying levels of crime rates against women and identifying potential factors contributing to these differences.
3. **To highlight the relationship between state/UT and crime type**: Investigating if certain states or UTs are more prone to specific types of crimes, such as dowry deaths or sexual assault.

4. **To inform policy and intervention measures**: Providing insights that could help policymakers design region-specific interventions and improve existing support systems for women affected by crime.

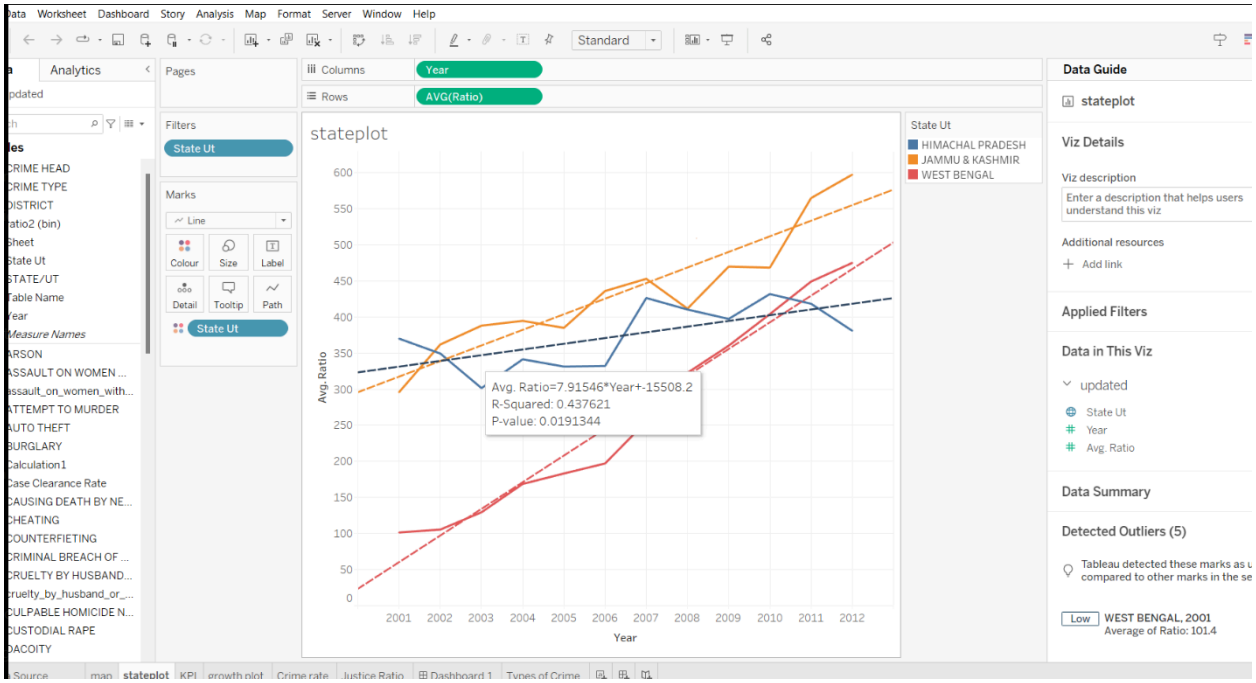
Choropleth map



This map is a **choropleth map** created in Tableau. A choropleth map uses color shading to represent data values for predefined geographic areas, like states or districts in this case. Here's how it works in your map:

- **Color Gradient**: Each state or union territory in India is shaded in a color that represents the average value of "ratio/100"(where ratio = total value of women crime in that state/total population of women in that state). Darker shades indicate higher values, while lighter shades indicate lower values.
- **Data Interpretation**: By looking at the color differences across states, you can quickly identify which regions have higher or lower values of the calculated metric. Dark blue areas, for instance, have the highest average values, while lighter areas have lower values.
- **Purpose**: Choropleth maps like this are useful for showing how a metric (like crime rates) varies across geographic regions, making it easier to identify patterns or trends across the country.

StatePlot :



The chart displays the average crime ratio trends over time for three states: Himachal Pradesh, Jammu & Kashmir, and West Bengal. The x-axis represents the years, while the y-axis shows the average crime ratio.

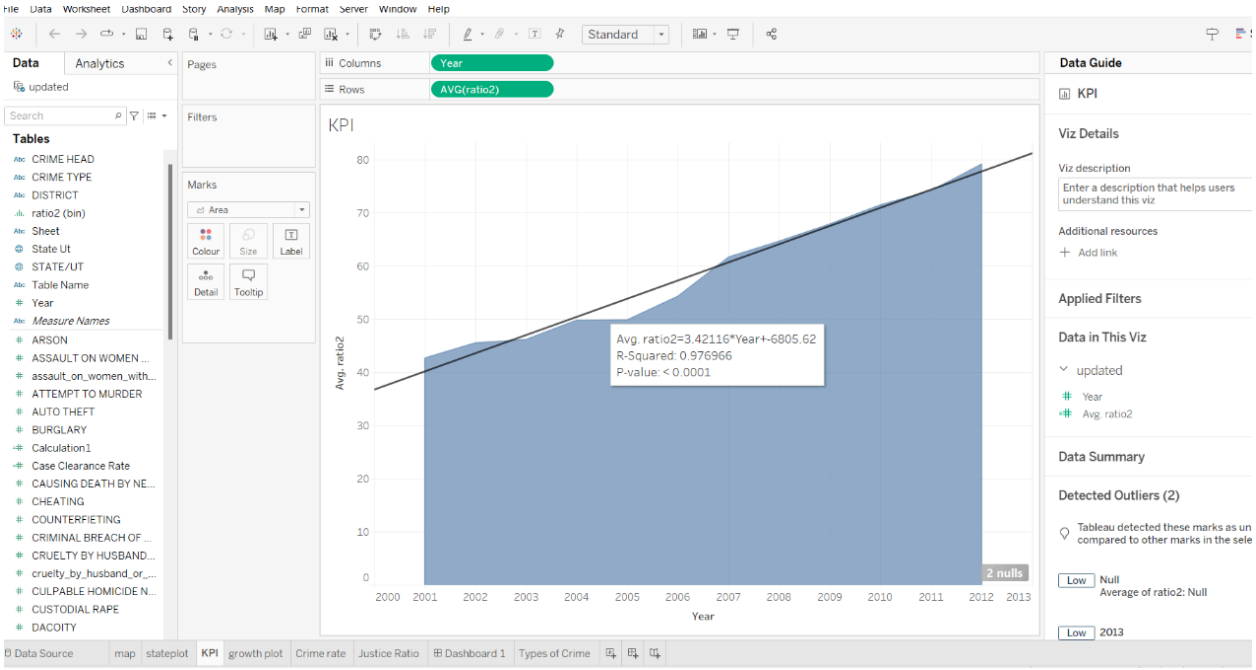
Observations:

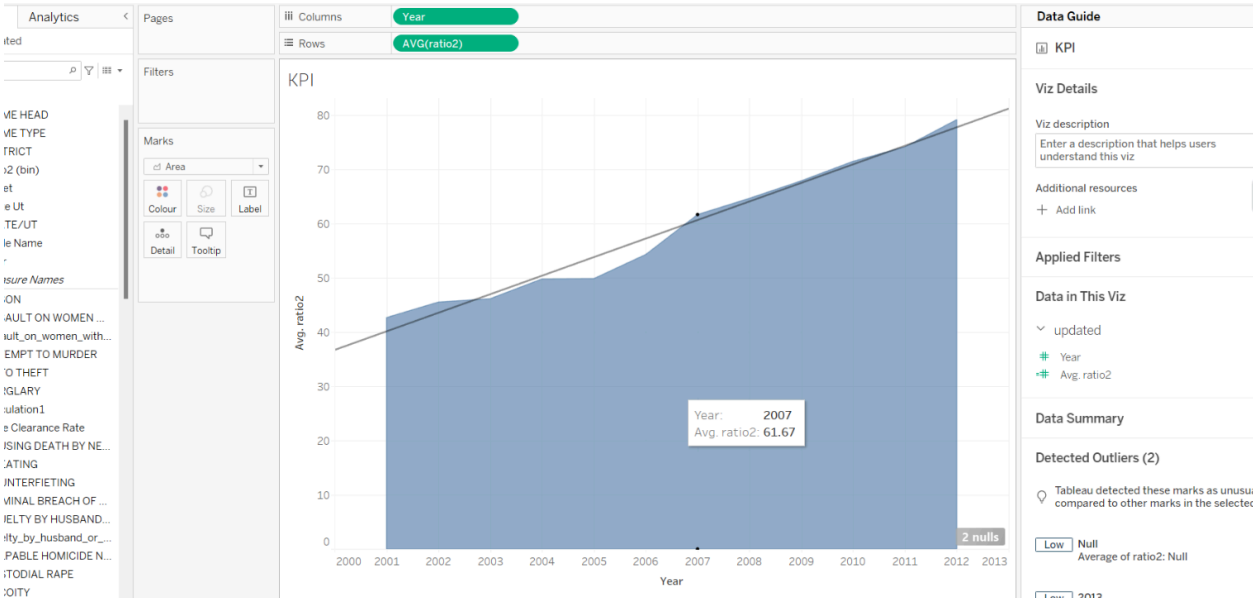
1. **Himachal Pradesh** (Blue Line):
    - The crime ratio in Himachal Pradesh shows a steady increase over the years but at a slower rate compared to the other states.
    - It starts at a relatively low value in the early 2000s and gradually rises.
2. **Jammu & Kashmir** (Orange Line):
    - The crime ratio in Jammu & Kashmir has a more prominent upward trend, with fluctuations in some years.
    - The increase in crime ratio is relatively consistent, indicating a steady rise in reported crimes over the period.
3. **West Bengal** (Red Line):
    - West Bengal's crime ratio starts lower in the early years but rises sharply and eventually surpasses the other states around 2010.
    - The steep rise indicates a rapid increase in the crime ratio, making it the highest among the three states by the end of the timeline.

Key Insights:

- **Trend Differences**: While all three states show an upward trend, West Bengal's rate of increase is the steepest, suggesting a significant rise in crime rates over time.
- **Outlier Detection**: Tableau has flagged some years (like West Bengal in 2001) as outliers, which could indicate unusual drops or spikes in crime for those specific years.

KPI


File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Data
Analytics
updated
Tables
CRIME HEAD
CRIME TYPE
DISTRICT
ratio2 (bin)
Sheet
State Ut
STATE/UT
Table Name
Year
Measure Names
ARSON
ASSAULT ON WOMEN ...
assault_on_women_with...
ATTEMPT TO MURDER
AUTO THEFT
BURGLARY
Calculation1
Case Clearance Rate
CAUSING DEATH BY NE...
CHEATING
COUNTERFIETING
CRIMINAL BREACH OF ...
CRUELTY BY HUSBAND...
cruelty_by_husband_or_...
CULPABLE HOMICIDE N...
CUSTODIAL RAPE
DACOITY
Pages
Filters
Marks
Area
Colour
Size
Label
Detail
Tooltip
Columns
Year
Rows
AVG(ratio2)
KPI
Avg. ratio2=3.42116*Year+-6805.62
R-Squared: 0.976966
P-value: < 0.0001
Avg. ratio2
Year
2 nulls
Data Guide
KPI
Viz Details
Viz description
Enter a description that helps users understand this viz
Additional resources
Add link
Applied Filters
Data in This Viz
updated
Year
Avg. ratio2
Data Summary
Detected Outliers (2)
Tableau detected these marks as un
compared to other marks in the sele
Low
Null
Average of ratio2: Null
Low
2013
Data Source
map
stateplot
KPI
growth plot
Crime rate
Justice Ratio
Dashboard 1
Types of Crime


Analytics
Pages
Filters
Marks
Area
Colour
Size
Label
Detail
Tooltip
Columns
Year
Rows
AVG(ratio2)
KPI
Year: 2007
Avg. ratio2: 61.67
Avg. ratio2
Year
2 nulls
Data Guide
KPI
Viz Details
Viz description
Enter a description that helps users understand this viz
Additional resources
Add link
Applied Filters
Data in This Viz
updated
Year
Avg. ratio2
Data Summary
Detected Outliers (2)
Tableau detected these marks as unusu
compared to other marks in the selecte
Low
Null
Average of ratio2: Null

Overview

Tableau visualization with an area chart displaying the trend in "Avg. ratio2" over the years. A linear trend line is added to depict the overall direction of the data, along with statistical details like the equation, R-squared value, and p-value .

Key Insights

1. Trend Line & Equation :

- The first image includes a tooltip with the trend line equation: Avg. ratio2=3.42116×Year−6805.62\text{Avg. ratio2} = 3.42116 \times \text{Year} - 6805.62Avg. ratio2=3.42116×Year−6805.62
- This equation shows that "Avg. ratio2" increases over time, with a slope of approximately 3.42, indicating a steady year-over-year rise.
- The R-squared value is **0.976966**, indicating a very strong fit between the trend line and the actual data points. This suggests that the linear model explains a high percentage of the variance in "Avg. ratio2."
- The p-value is **<0.0001**, suggesting that the trend is statistically significant.

2. Yearly Data and Outliers:

- The data rises gradually, with some fluctuations, until around 2007.
- In **2007**, "Avg. ratio2" is highlighted with a value of **61.67**, marking a notable data point along the trend.
- Tableau has detected **two outliers** (as shown on the right under "Detected Outliers" in both images):
    - One is for the year **2013**, with a significantly low or null value, which could represent missing or unexpected data for that year.
    - The other outlier is labeled as "Null," indicating possible missing data in the dataset.

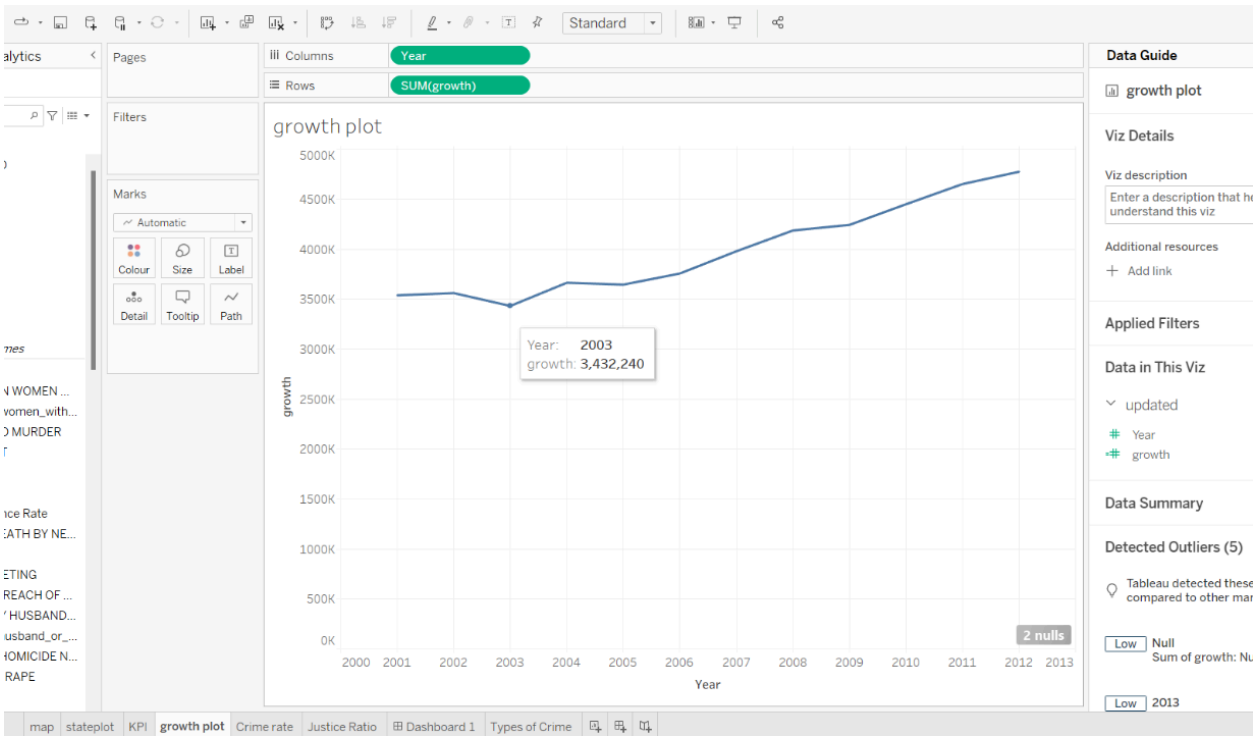
3. Shaded Area:

- The area under the trend represents the accumulated "Avg. ratio2" values over time. The increasing shaded area visually reinforces the upward trend.

4. Data Fluctuations:

- From 2000 to approximately 2007, "Avg. ratio2" shows a relatively smooth increase with some minor dips.
- After 2007, the values continue to rise steadily but fluctuate slightly closer to the trend line.

Growth Plot :



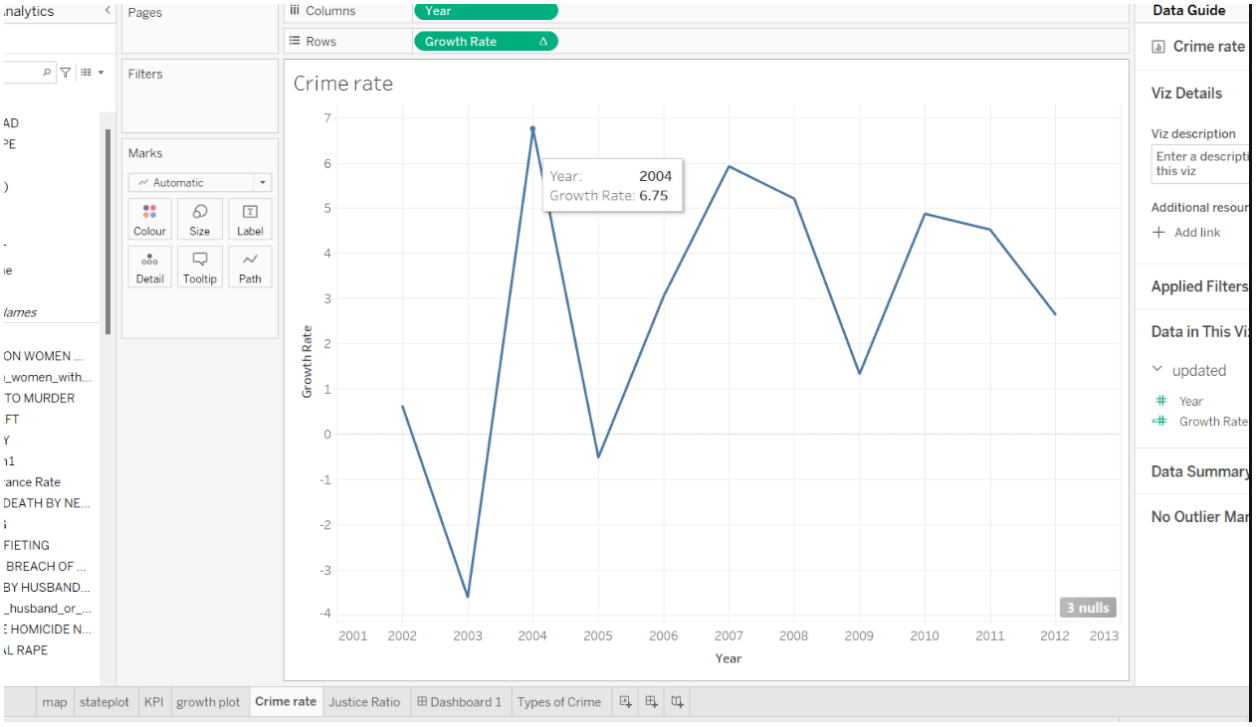
It shows a "growth plot" in Tableau, which tracks the "SUM(growth)" metric over the years.

Explanation of the Growth Plot

1. **Overall Trend**:
    - The line chart displays the sum of "growth" over time, from 2000 to 2012.

    - There is a generally upward trend, indicating that "growth" has been increasing over the years.
2. **Yearly Fluctuations**:
    - In **2003**, the tooltip shows that the "growth" value was **3,432,240**. This point represents a dip or decrease compared to previous and subsequent years, where the growth otherwise rises steadily.
3. **Notable Increases**:
    - From 2004 onward, the plot shows a consistent increase, with some years experiencing steeper inclines than others, indicating periodic accelerations in growth.
4. **Outliers**:
    - Similar to the previous charts, **Tableau has detected two outliers**:
        - One is for the year **2013**, where the value of "growth" might be unusually low or null, suggesting missing or unexpected data.
        - Another outlier is labeled "Null," indicating there might be missing data points in the dataset.

Crime Rate :



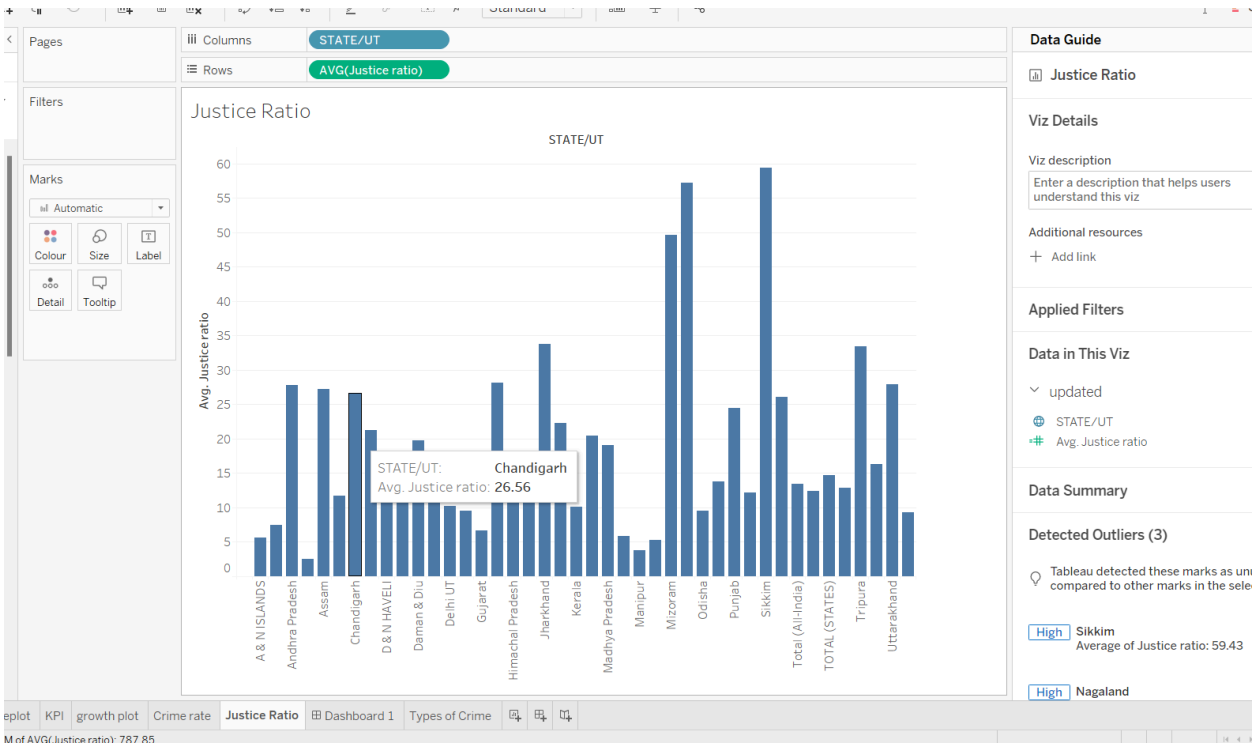
It shows a "Crime Rate" line chart in Tableau, tracking the **Growth Rate** of crime over time from 2001 to 2013.

Explanation of the Crime Rate Plot

1. **Overall Trend**:
   - The line chart displays fluctuations in crime growth rates over the years, showing both positive and negative values. This indicates varying growth and decline in crime rates from year to year rather than a steady trend.
2. **Significant Points**:
   - **2002**: There is a significant dip with a negative growth rate, indicating a decline in crime that year.
   - **2004**: The tooltip indicates a peak in crime growth rate with a value of **6.75**, showing a significant rise in crime growth that year.
   - **2007, 2009, and 2011**: These years show other peaks, though not as high as in 2004, suggesting periodic increases in the crime growth rate.
   - **2003 and other years**: Show declines in growth rates, reflecting periods when crime rates might have decreased or growth was slower.

3. **Detected Nulls**:
    - The chart notes **3 nulls** in the lower right corner, indicating that data is missing for certain years, which might impact the continuity of the analysis.

Justice Ratio :



The presented visualization is a **bar chart** created using Tableau to analyze the **average Justice Ratio** across various Indian states and Union Territories (UTs). This metric likely reflects the efficiency of the justice delivery system in each region. The analysis provides valuable insights into regional disparities in justice administration.

Key Observations:

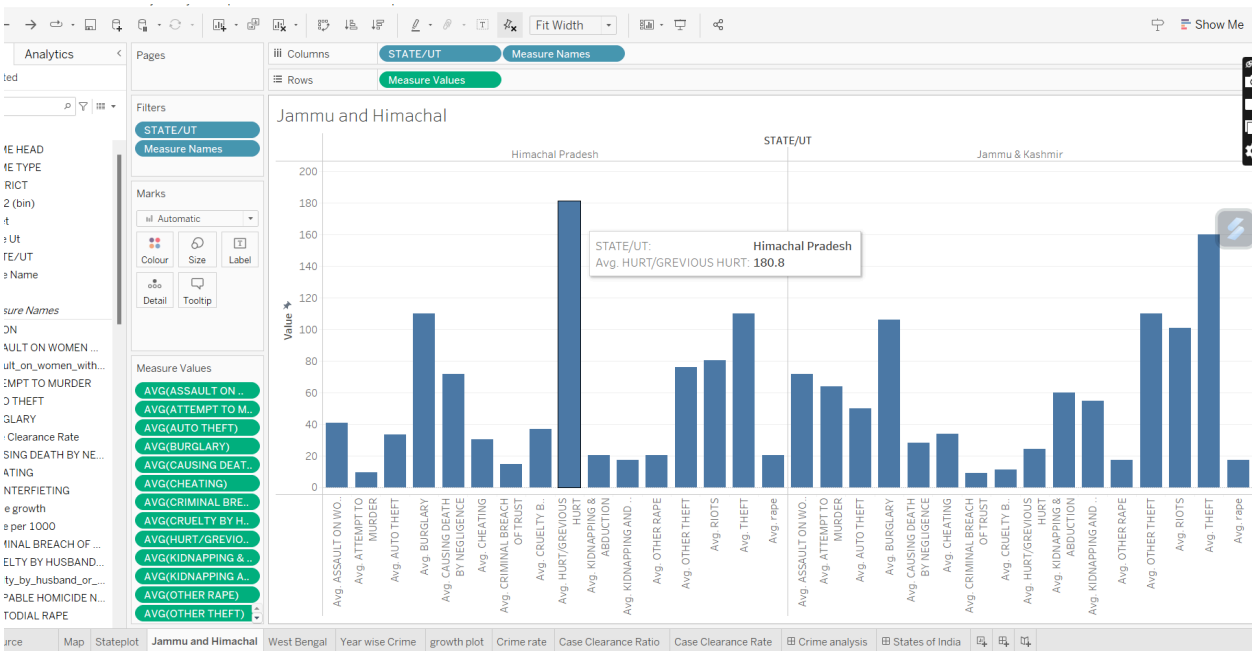
1. **Chart Structure:**
    - **X-axis:** Represents the different Indian states and UTs, including regions such as Assam, Chandigarh, Kerala, Sikkim, and others.

- **Y-axis:** Indicates the **average Justice Ratio**, a numerical value that measures the performance or efficiency of the judicial system in each state/UT.
- **Data Points:** Each bar represents the average Justice Ratio for a specific state or UT. For instance, the average Justice Ratio for **Chandigarh** is **26.56**.

2. **Outlier Detection:**
   - Tableau has identified **three significant outliers** with notably high Justice Ratios:
     - **Sikkim** with an average Justice Ratio of **59.43**.
     - **Nagaland** with an average Justice Ratio of **57.27**.
     - **Mizoram** with an average Justice Ratio of **49.68 .**
   - These outliers indicate states where the Justice Ratio is significantly higher than the overall average, suggesting a relatively more efficient judicial process in these regions.
3. **Comparative Analysis:**
   - There is a noticeable variance in the Justice Ratios across different states and UTs. While some regions, such as **Sikkim** and **Nagaland**, exhibit higher Justice Ratios, others show lower values, indicating potential areas for improvement in judicial efficiency.
   - The visualization highlights differences that may be influenced by factors such as regional governance, resource allocation, case load, and infrastructure of the legal system.

State Analysis :

1)Jammu and Himachal



The provided Tableau visualization presents a comprehensive analysis of various crime metrics across the Indian states of Himachal Pradesh and Jammu & Kashmir. This bar chart aims to offer insights into the prevalence and distribution of different types of crimes in these regions, which can be instrumental for law enforcement and policy-making.

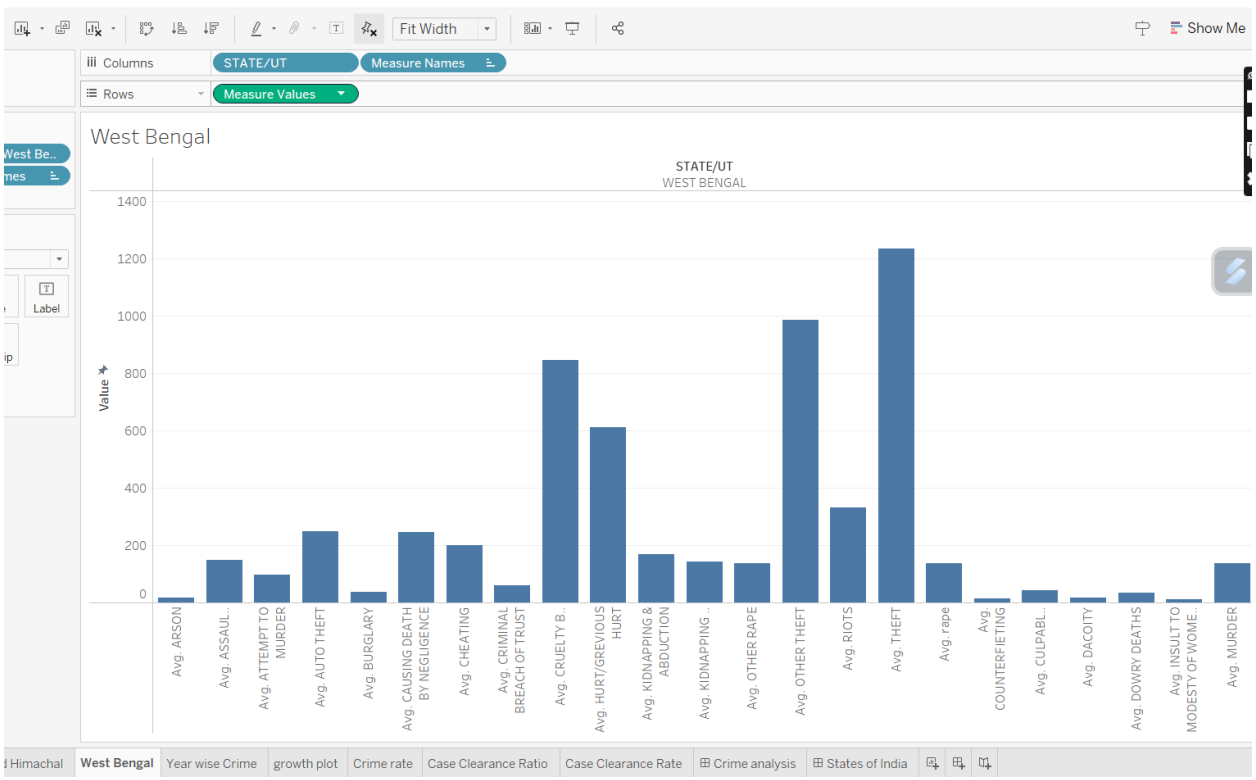
Himachal Pradesh shows an average of 180.8 incidents for the crime category Hurt/Grievous Hurt.
This metric indicates a relatively high incidence of this specific violent crime type in the state, highlighting an area of concern for law enforcement.

Comparative Analysis:

- The chart illustrates notable variations in the average values of different crime types between the two states:

- **Hurt/Grievous Hurt** appears to have a particularly high average in Himachal Pradesh, indicating a significant frequency of violent incidents.
- **Theft**, including both auto theft and other theft types, also shows relatively higher averages, suggesting common property crimes in the dataset.
- Lower average values for crimes like **Cheating** and **Burglary** may indicate less frequent reporting or lower incidence rates for these categories compared to violent crimes or thefts.

2) West Bengal



The provided Tableau visualization offers an in-depth analysis of various crime metrics for the state of **West Bengal**. Below is a detailed explanation suitable for a formal report:

Overview of the Visualization

The bar chart illustrates the average incidence rates of different types of crimes in **West Bengal**, providing a comparative view across various crime categories. The data representation is designed to help identify the most prevalent types of criminal activities in the state.

Key Observations:

- **Theft:** The highest bar in the chart corresponds to the crime category **Theft**, with an average incidence value of **1,233**. This suggests that theft is a significant issue in West Bengal, making it the most frequently reported crime in this dataset.
- **Cruelty by Husband or Relatives:** This category also shows a high average, indicating prevalent domestic violence cases. This metric highlights a pressing social issue in the region that may require focused intervention and support mechanisms.
- **Hurt/Grievous Hurt:** The average incidence rate for hurt-related crimes is also relatively high, reflecting a substantial number of reported cases involving physical injuries.
- **Auto Theft and Burglary:** Both these property crimes have lower averages compared to theft but still represent notable issues, showing moderate rates of occurrence.

Questions and Analysis:

1. What Factors Could Explain the Trends in Crime Growth?

Trends in crime growth often reflect a mix of legislative changes, public awareness, socio-economic factors, media influence, and judicial effectiveness:

1. **Legislative and Policy Shifts**: New or stricter laws, especially around issues like assault and domestic violence, can lead to short-term increases in reported cases as law enforcement attention and public trust rise.

2. **Awareness Campaigns**: Initiatives from governments and NGOs to highlight issues like domestic violence or harassment often encourage more victims to report crimes, making data more reflective of actual incidents.
3. **Economic Factors**: Economic stress, poverty, and inequality can contribute to higher crime rates, as financial pressures exacerbate domestic and societal tensions.
4. **Judicial and Police Efficiency**: Effective law enforcement and quick judicial action tend to deter crime, whereas inefficiency or corruption can lead to persistently high rates, as offenders feel emboldened.

2.How does the crime growth rate fluctuate each year, and what does it reveal about crime trends?

- **Identifying Specific Events or Policies**: Years with peaks, such as 2004 with a high growth rate of 6.75%, might correspond to specific incidents that increased public awareness or government focus on crime. Understanding such events allows us to contextualize crime rates and identify if similar interventions could be applied in other years.
- **Interventions and Temporary Drops**: Sharp declines, like those seen in 2003, may indicate successful policy measures or campaigns that temporarily reduced crime rates. However, if these drops are not sustained, it could imply that the intervention was either too short-lived or not effectively institutionalized.
- **Comparing Against External Factors**: Economic crises, significant political events, or natural disasters could correlate with drops or spikes in reported cases. For instance, in times of national emergencies, resources may shift away from crime prevention, affecting reporting and enforcement.

3.Are there any cyclical or repeating patterns in the crime growth rate?

- **Seasonal Variations**: In some cases, crime rates might increase during particular seasons due to social events or festivals where women are at greater risk (e.g., high travel periods or public gatherings). Analyzing monthly or quarterly data, if available, could reveal these patterns and inform seasonal prevention campaigns.

- **Impact of Election Cycles and Government Transitions**: In regions where new governments often implement or reform laws, cycles in crime rates might be seen around election periods or major administrative changes. Identifying such trends can indicate if enforcement is politically motivated or inconsistent.
- **Long-Term Trends**: If a pattern emerges over multiple years, it may point to structural or cultural issues that persist regardless of interventions. For instance, a pattern of increasing crime every few years could suggest a lapse in enforcement efforts or a need for sustained educational campaigns.

4.What might explain the sharp decline in certain years, such as 2003?

- **Effect of Temporary Policies or Crackdowns**: A decline in reported cases could indicate a period of heightened law enforcement activity or a crackdown on specific crimes, such as human trafficking or dowry-related violence. It's essential to evaluate if such measures were one-off campaigns or long-term reforms.
- **Changes in Reporting Mechanisms**: The implementation of helplines or women's police stations may initially decrease reported crimes if victims are uncertain about the new process. However, over time, these mechanisms typically increase reporting as trust builds.
- **Economic or Social Events**: Major events like recessions may reduce crime rates temporarily, as economic stress impacts reporting or law enforcement resource allocation. For instance, some regions might deprioritize women's safety in times of broader crisis, impacting reported cases.

5. Are There States with Consistently Low or High Crime Rates Over the Years?

- Socio-Cultural Norms and Attitudes: States with lower crime rates might have stronger cultural norms that discourage crimes against women, or they might have lower reporting rates due to cultural stigma. High-crime states may reflect cultural practices (like dowry), economic issues, or social inequality that exacerbate women's vulnerability.
- Legal and Policy Differences: Regions with well-funded police forces, better judicial infrastructure, or dedicated resources for women's protection often

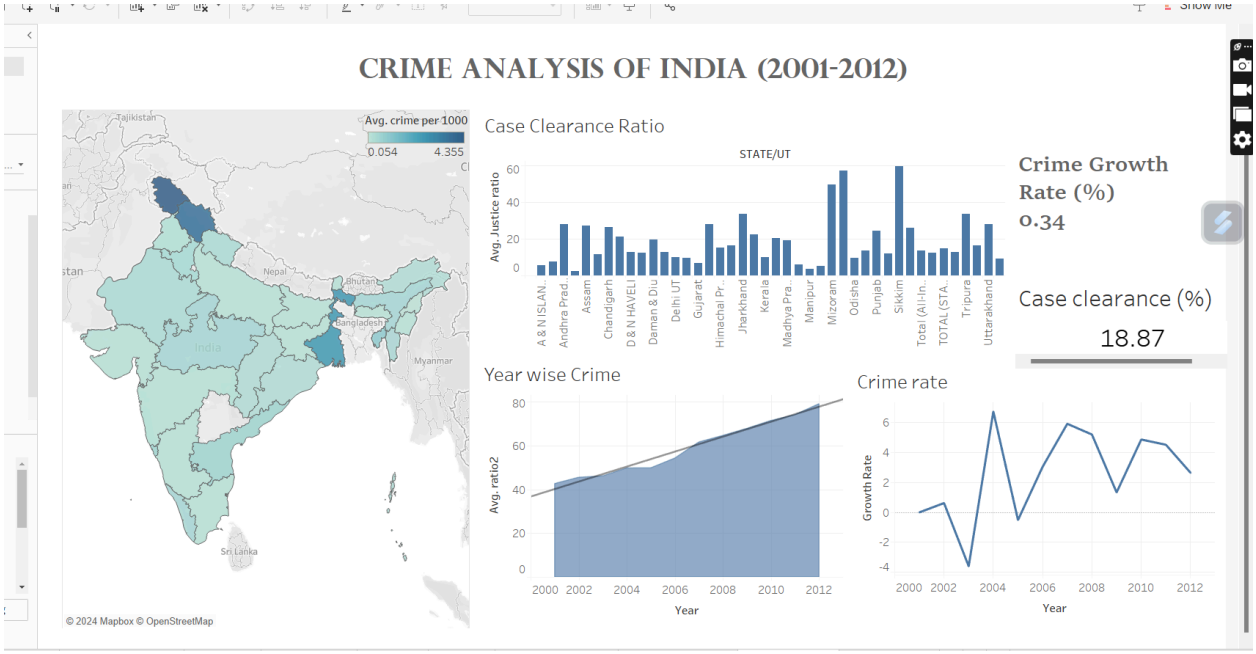
have consistently lower crime rates. High-crime regions might lack such resources or have weaker law enforcement.

- Urban vs. Rural Divide: Some regions with major urban centers may report higher crime rates due to higher population density, greater anonymity, or increased awareness and reporting. Rural areas, despite potentially high crime levels, may underreport due to limited access to justice or social pressure.

How does the distribution of crime types vary by region?

- **Urban vs. Rural Factors**: Urban centers may report higher rates of sexual harassment and trafficking due to migration, anonymity, and diverse populations. Rural areas, however, might have higher rates of domestic violence or dowry deaths due to traditional beliefs and economic pressures.
- **Border Regions and Trafficking**: States bordering other countries (e.g., West Bengal, Assam) may report higher incidents of trafficking due to porous borders. These regions may need stricter border control and support for trafficked individuals.
- **Intervention Potential**: Regions with a high prevalence of certain crimes could be targeted with specific programs. For example, high dowry death states might benefit from stricter dowry laws and public education campaigns, whereas states with high rates of sexual harassment might focus on workplace safety regulations.

DashBoard :


CRIME ANALYSIS OF INDIA (2001-2012)
Avg. crime per 1000
0.054
4.355
Case Clearance Ratio
STATE/UT
Avg. Justice ratio
Crime Growth Rate (%)
0.34
Case clearance (%)
18.87
Year wise Crime
Avg. ratio2
Year
Crime rate
Growth Rate
Year
© 2024 Mapbox © OpenStreetMap

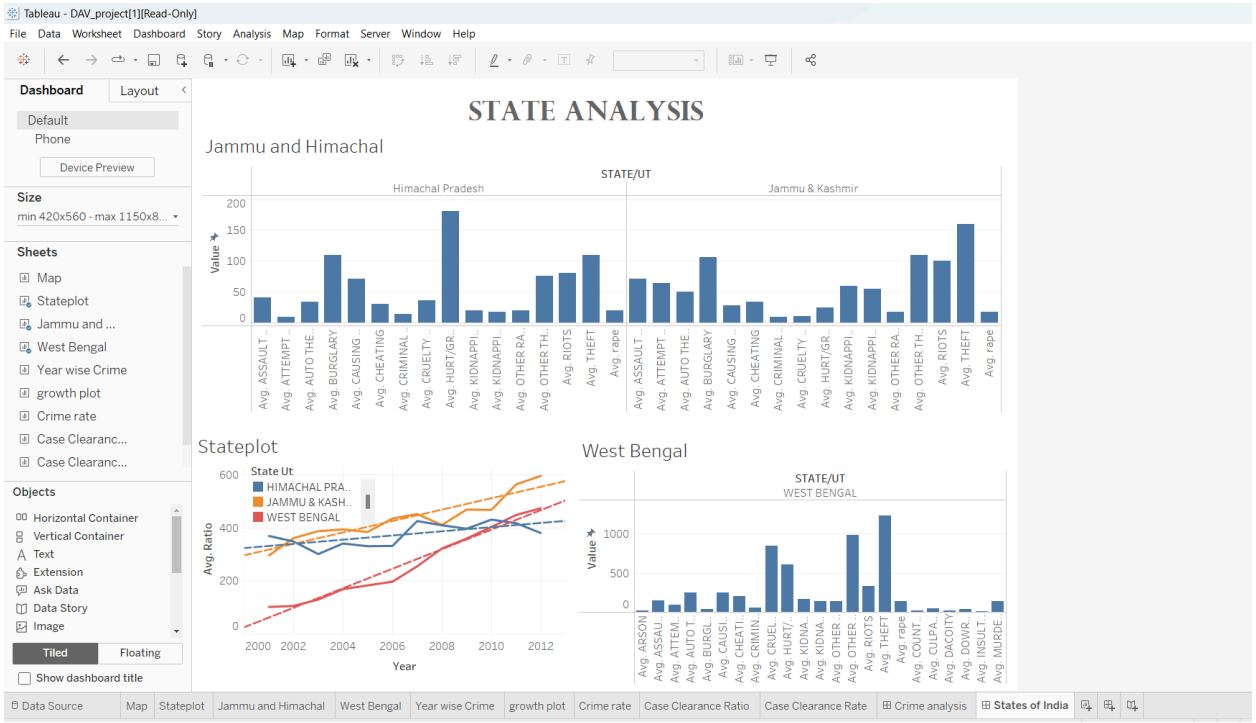


Tableau - DAV_project[1][Read-Only]
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Dashboard
Layout
Default
Phone
Device Preview
Size
min 420x560 - max 1150x8...
Sheets
Map
Stateplot
Jammu and ...
West Bengal
Year wise Crime
growth plot
Crime rate
Case Clearanc...
Case Clearanc...
Objects
Horizontal Container
Vertical Container
Text
Extension
Ask Data
Data Story
Image
Tiled
Floating
Show dashboard title
STATE ANALYSIS
Jammu and Himachal
STATE/UT
Himachal Pradesh
Jammu & Kashmir
Value
Stateplot
State Ut
HIMACHAL PRA..
JAMMU & KASH..
WEST BENGAL
Avg. Ratio
Year
West Bengal
STATE/UT
WEST BENGAL
Value
Data Source
Map
Stateplot
Jammu and Himachal
West Bengal
Year wise Crime
growth plot
Crime rate
Case Clearance Ratio
Case Clearance Rate
Crime analysis
States of India